AECL Advanced ElectroChemistry Lab

A Study on the Two-Step Electrochemical Fabrication of CFTS (Cu_2FeSnS_4) Thin-Film Solar Cells via Electrodeposition and Sulfurization

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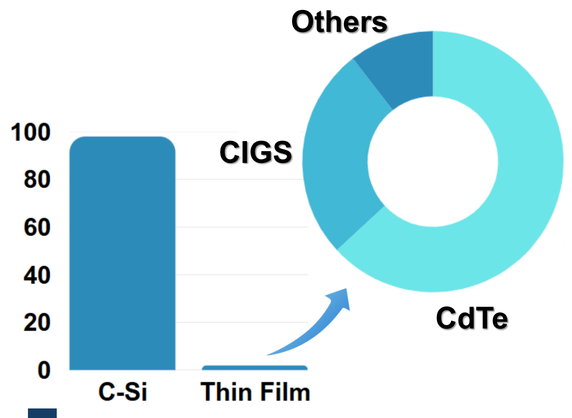
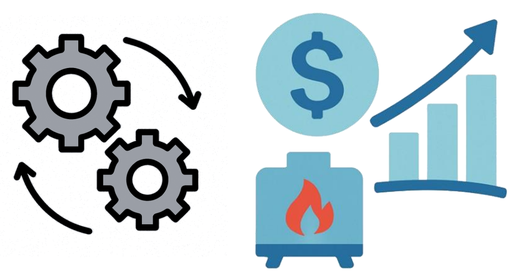
Introduction

Market Share by Solar Cell



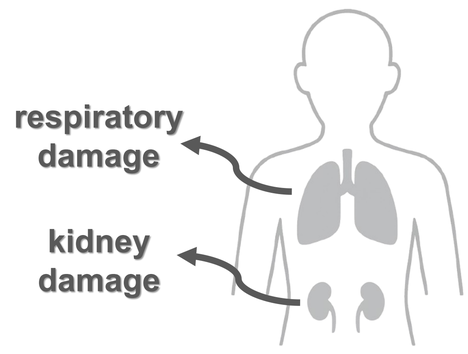
C-Si

- High process complexity
- High production cost



CdTe

- Toxicity of Cd

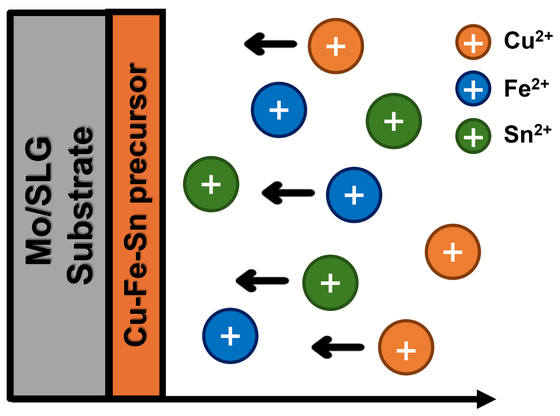


CFTS

- Non-toxic components
- Low production cost

Experiment

Step 1 Electrodeposition



Electrolyte (0.2 L)

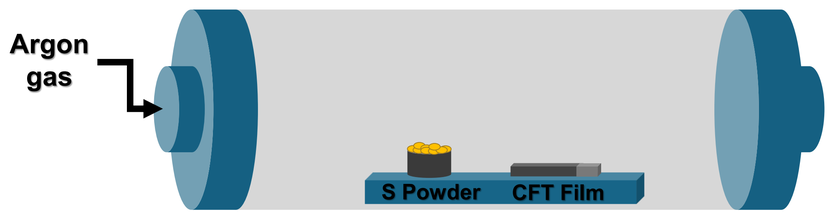
- Raw materials :
  $FeSO_4 \cdot 7H_2O$ (0.01 M, 0.03 M, 0.05 M)
  $SnSO_4 \cdot 4H_2O$ (0.015 M)
  $CuSO_4$ (0.02 M)
- Complexing agents :
  Trisodium citrate (0.1 M)
  EDTA (0.05 M)
- pH : 5

Electrode

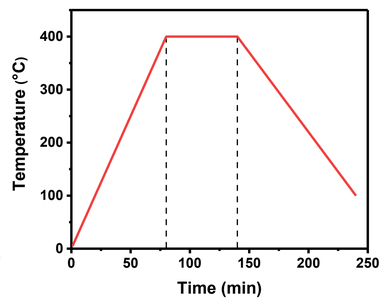
- Working electrode: Mo/ SLG(Soda lime glass)
- Counter electrode: Pt Coil
- Reference electrode: Ag/AgCl

Step 2 Sulfurization

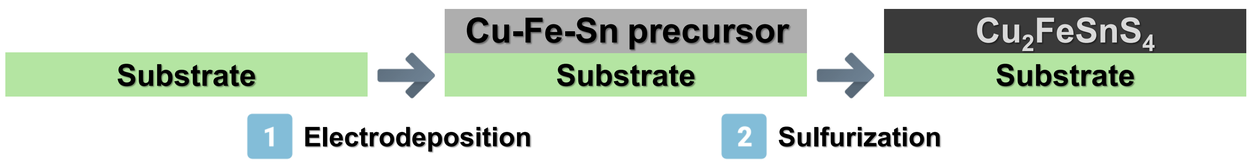
- Argon atmosphere (10 sccm flow rate)
- 400 ℃ for 1 hour (heating rate: 5 ℃ /min)



✓ Sulfurization conditions



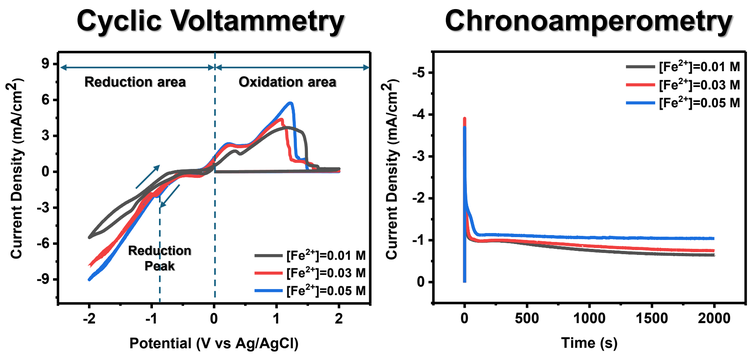
The CFT (Cu–Fe–Sn) precursor film was sulfurized, resulting in the formation of a CFTS layer.



Results & Discussion

Step 1: Electrodeposition

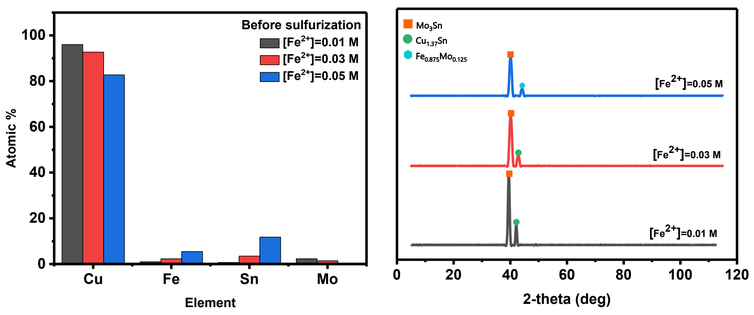
Effect of Fe^{2+} Concentration on Electrodeposition Behavior



Experimental Conditions

- Specimen : SLG / Mo (200 nm) Mo deposited by sputtering
- Potential : - 0.9 V (from CV reduction peak)
- Time : 2000 s
- Temperature : RT

Chemical Composition and Phase Analysis of CFT Precursors



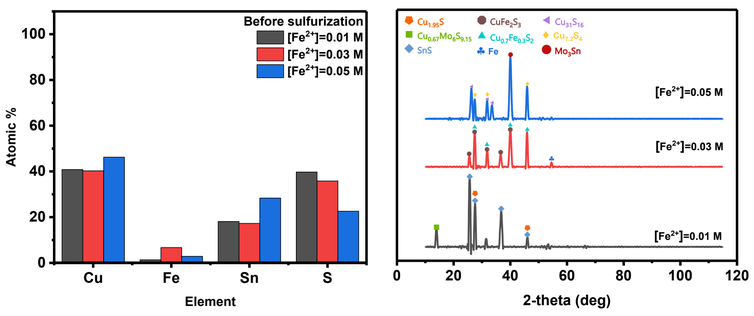
Discussion

The co-deposited precursor exhibited a Fe–Sn-poor composition

Likely due to insufficient Fe and Sn incorporation during electrodeposition

Step 2: Sulfurization

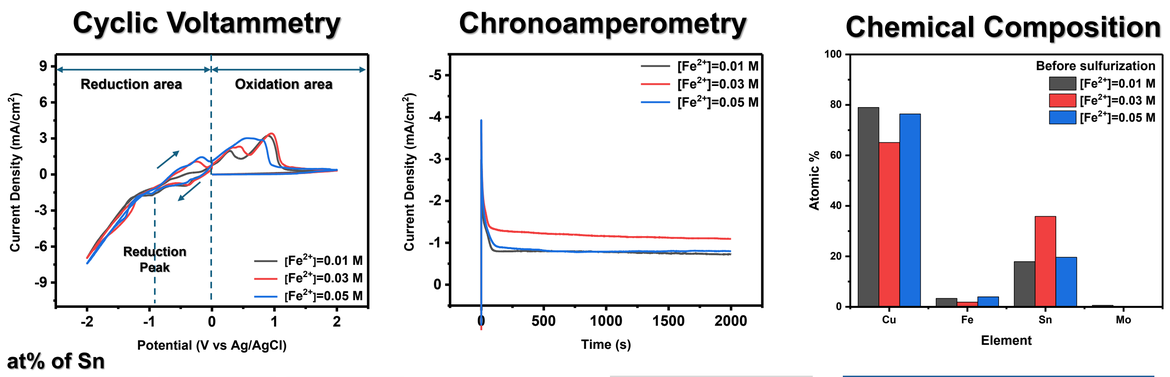
Chemical Composition and Phase Analysis of CFTS film



Discussion

Insufficient Fe–Sn incorporation in the precursor likely suppressed the formation of stoichiometric CFTS, leading to the formation of secondary phases

Effect of EDTA Removal in the Electrolyte

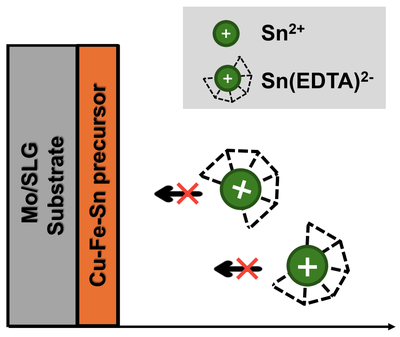


at% of Sn

[Fe^{2+}]	EDTA (O)	EDTA (X)
0.01 M	0.39%	17.90%
0.03 M	3.49%	35.82%
0.05 M	11.79%	19.64%

at% of Fe

[Fe^{2+}]	EDTA (O)	EDTA (X)
0.01 M	1.01%	3.33%
0.03 M	2.29%	1.86%
0.05 M	5.47%	3.96%



Discussion

EDTA removal enhanced Sn incorporation by preventing stable Sn–EDTA complex formation; however, Fe incorporation remained insufficient, necessitating further electrochemical study.

Conclusion

- The CFT precursor was electrodeposited at the reduction potential of -0.9 V, as determined from the CV measurement
- Electrodeposited CFT precursors showed insufficient Fe–Sn incorporation, hindering the formation of stoichiometric CFTS films
- Deposition under EDTA-free conditions effectively resolved Sn deficiency by eliminating stable Sn–EDTA complexes.
- However, unresolved Fe deficiency suggests that further electrochemical optimization is required.